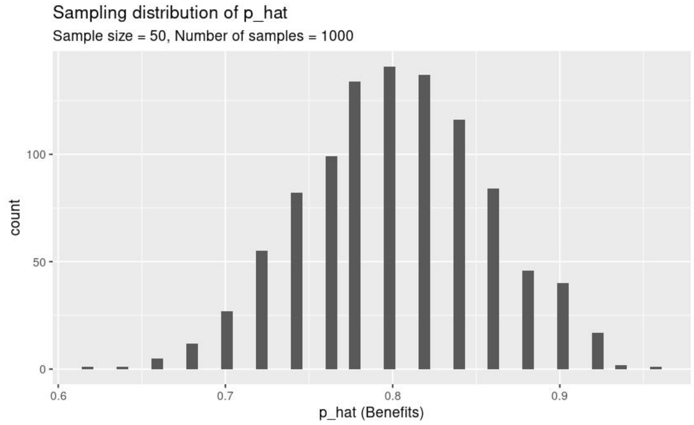
General features of sampling distributions

How is a sampling distribution constructed?

- Select a simple random sample of size n from the population. Compute the sampling statistic based on that sample. Plot that sample statistic on a dotplot.

- Repeat this process for **every possible sample from the population**. Typically this is too many for even a computer: for just of 50 cases drawn from a population of 1000, there are $\binom{1000}{50}$ possible samples, which is about 9.4x10^84, or more than the number of atoms in the known universe!

- So in practice we just do it many, many times (at least 1000), and we don't worry about if we're accidentally repeating any sample, since that is very unlikely to happen. The resulting dotplot is called a sampling distribution for that sample statistic.



General features of sampling distributions

As long as certain conditions are met about our sample (future excitement: read about the *Central Limit Theorem*), sampling distributions will always be

- Bell shaped
- Centered at the value of the *population parameter* that corresponds to the sample statistic you're plotting.

Remarkably, this happens regardless of the shape of the original population's distribution from which you're sampling!

Relationship between Sample Size and Spread/Variability

What happens to the sampling distribution as the sample size increases?

- (Shape) As the sample size increases, the sampling distribution becomes more and more symmetric and mound-shaped.
- (Center) As the sample size increases, the center of the sampling distribution tends to get closer and closer to the population parameter.
- (Spread) As the sample size increases, the *standard error** decreases. This means that as the sample size increases, the sample statistics tend to be more greatly concentrated about the population parameter.

  In other words: a **larger sample size** is more likely to produce a sample statistic that is closer to the population parameter. But that doesn't mean larger is always better: a small study that we can afford to actually carry out is better than a large study we can't afford!

***Important definition:** The standard error is the standard deviation of the sampling distribution.